

Student name:

CCNA v7.0 –SEMETER 2
FINAL SKILL EXAM
Duration: 1 hour 45 min

1. Object

- Build the network topology according to Device configuration and Physical connection tables.
- Configure LACP port-channel, VLAN, STP, HSRP on routers/switched according to Planning section.
- Configure DHCP service and WLAN as requirements.
- Configure OSPF routing protocol and static route to ensure connectivity between networks.
- Apply port-security and some security features on switches.

2. Device configuration

Site	Location/Function	Host name	Model
Head Quarter	Datacenter/DC	HQ.GW01	Cisco CISCO2911/K9
Head Quarter	Datacenter/DC	DC.DS01	Cisco WS-C3560-24PS
Head Quarter	Datacenter/DC	DC.DS02	Cisco WS-C3560-24PS
Head Quarter	Datacenter/DC	DC.ACC01	Cisco WS-C2960-24TT
Head Quarter	Datacenter/DC	DC.ACC02	Cisco WS-C2960-24TT
Head Quarter	Datacenter/Office	OFF.DS01	Cisco WS-C3560-24PS
Head Quarter	Datacenter/Office	OFF.WLC01	Cisco WLC-3504
Head Quarter	Datacenter/DC	mb.com.vn	Server-PT
Head Quarter	Datacenter/DC	APP1	Server-PT
Head Quarter	Datacenter/DC	DB1	Server-PT
Head Quarter	Datacenter/DC	DB2	Server-PT
Head Quarter	Datacenter/Office	OFF.DHCP01	Server-PT
Head Quarter	Floor-1/Office	OFF.ACC-F1	Cisco WS-C2960-24TT
Head Quarter	Floor-1/Office	OFF.LAP-F1	LAP-PT
Head Quarter	Floor-2/Office	OFF.ACC-F2	Cisco WS-C2960-24TT
Head Quarter	Floor-2/Office	OFF.LAP-F2	LAP-PT
Head Quarter	Floor-1/Office	Accountant1	PC-PT
Head Quarter	Floor-1/Office	Counter1	PC-PT
Head Quarter	Floor-1/Office	Guest1	Laptop-PT
Head Quarter	Floor-2/Office	Accountant2	PC-PT
Head Quarter	Floor-2/Office	Counter2	PC-PT
Head Quarter	Floor-2/Office	Guest2	Laptop-PT
Branch	Office	BR.GW01	Cisco CISCO1941/K9
Branch	Office	BR.ACC01	Cisco WS-C2960-24TT
Branch	Office	BR.AP01	AccessPoint-PT

Site	Location/Function	Host name	Model
Branch	Office	Accountant3	PC-PT
Branch	Office	Counter3	PC-PT
Branch	Office	Guest3	Laptop-PT

3. Planning

3.1. Physical connection and port-channel

Host name	Interface	Port-channel	Port-channel	Interface	Host name	Note
HQ.GW01	S0/0/0			S0/0/0	FPT Telecom	
HQ.GW01	S0/0/1			S0/0/1	BR.GW01	
HQ.GW01	Gi0/0			Gi0/1	OFF.DS01	
HQ.GW01	Gi0/1			Gi0/1	DC.DS01	
HQ.GW01	Gi0/2			Gi0/1	DC.DS02	
DC.DS01	Fa0/23	Po6	Po6	Fa0/23	DC.DS02	Link Aggregation Protocol: LACP
DC.DS01	Fa0/24			Fa0/24	DC.DS02	
DC.DS01	Fa0/19	Po1	Po1	Fa0/19	DC.ACC01	Link Aggregation Protocol: LACP
DC.DS01	Fa0/20			Fa0/20	DC.ACC01	
DC.DS01	Fa0/21	Po2	Po1	Fa0/19	DC.ACC02	Link Aggregation Protocol: LACP
DC.DS01	Fa0/22			Fa0/20	DC.ACC02	
DC.DS02	Fa0/19	Po1	Po2	Fa0/21	DC.ACC01	Link Aggregation Protocol: LACP
DC.DS02	Fa0/20			Fa0/22	DC.ACC01	
DC.DS02	Fa0/21	Po2	Po2	Fa0/21	DC.ACC02	Link Aggregation Protocol: LACP
DC.DS02	Fa0/22			Fa0/22	DC.ACC02	
OFF.DS01	Fa0/21	Po1	Po1	Fa0/23	OFF.ACC-F1	Link Aggregation Protocol: LACP
OFF.DS01	Fa0/22			Fa0/24	OFF.ACC-F1	
OFF.DS01	Fa0/23	Po2	Po1	Fa0/23	OFF.ACC-F2	Link Aggregation Protocol: LACP
OFF.DS01	Fa0/24			Fa0/24	OFF.ACC-F2	
OFF.DS01	Fa0/19			NIC0	OFF.DHCP01	
OFF.DS01	Fa0/20			Gi1	OFF.WLC01	
DC.ACC01	Fa0/1			NIC0	mb.com.vn	
DC.ACC01	Fa0/11			NIC0	DB1	
DC.ACC02	Fa0/6			NIC0	APP1	
DC.ACC02	Fa0/11			NIC0	DB2	
OFF.ACC-F1	Fa0/22			Gi0	OFF.LAP-F1	
OFF.ACC-F1	Fa0/1			NIC0	Accountant1	
OFF.ACC-F1	Fa0/11			NIC0	Counter1	
OFF.ACC-F2	Fa0/22			Gi0	OFF.LAP-F2	
OFF.ACC-F2	Fa0/1			NIC0	Accountant2	
OFF.ACC-F2	Fa0/11			NIC0	Counter2	
BR.GW01	S0/0/0			S0/0/1	FPT Telecom	
BR.GW01	Gi0/0			Gi0/1	BR.ACC01	

Host name	Interface	Port-channel	Port-channel	Interface	Host name	Note
BR.ACC01	Fa0/22			Port0	BR.AP01	
BR.ACC01	Fa0/1			NIC0	Accountant3	
BR.ACC01	Fa0/11			NIC0	Counter3	

3.2. VLAN and Spanning-Tree

Site	VLAN ID	VLAN Name	STP Root primary	STP Root Secondary	Access devices
Head Quarter	110	WEB server	DC.DS01	DC.DS02	mb.com.vn
Head Quarter	120	APP server	DC.DS01	DC.DS02	APP1
Head Quarter	130	DB server	DC.DS02	DC.DS01	DB1, DB2
Head Quarter	190	Management	DC.DS02	DC.DS01	-
Head Quarter	210	Accountant	OFF.DS01	-	Accountant1, Accountant2
Head Quarter	220	Counter	OFF.DS01	-	Counter1, Counter2
Head Quarter	230	Guest	OFF.DS01	-	OFF.WLC01, OFF.LAP-F1, OFF.LAP-F2, Guest1, Guest2
Branch	810	Accountant	-	-	Accountant3
Branch	820	Counter	-	-	Counter3
Branch	830	Guest	-	-	BR.AP01, Guest3

- All sites should declare a VLAN ID 998 for native VLAN and VLAN ID 999 for not in-used port (black hole VLAN).
- All trunk links should be changed native VLAN to VLAN 998.

3.3. IP

Site	Link/Network	IP Network	Note
-	FPT LAN	118.69.79.0/24	
-	HQ.GW01 to FPT Telecom	118.69.255.0/30	Primary link
-	BR.GW01 to FPT Telecom	118.69.255.4/30	Backup link
-	HQ.GW01 to BR.GW01	118.69.255.8/30	
Head Quarter	HQ.GW01 to DC.DS01	172.23.255.0/30	
Head Quarter	HQ.GW01 to DC.DS02	172.23.255.4/30	
Head Quarter	HQ.GW01 to OFF.DS01	172.23.255.8/30	
Head Quarter	VLAN 110	172.16.110.0/24	
Head Quarter	VLAN 120	172.16.120.0/24	
Head Quarter	VLAN 130	172.16.130.0/24	
Head Quarter	VLAN 190	172.16.190.0/24	
Head Quarter	OFF.DS01 DHCP LAN	172.17.10.0/24	
Head Quarter	VLAN 210	172.17.210.0/24	
Head Quarter	VLAN 220	172.17.220.0/24	
Head Quarter	VLAN 230	172.17.230.0/24	

Site	Link/Network	IP Network	Note
Branch	VLAN 810	172.28.81.0/24	
Branch	VLAN 820	172.28.82.0/24	
Branch	VLAN 830	172.28.83.0/24	

3.4. HSRP

VLAN ID	Network	HSRP Configuration				
		Version	Group	VIP	Active router	Standby router
110	172.16.110.0/24	2	110	172.16.110.1	DC.DS01	DC.DS02
120	172.16.120.0/24	2	120	172.16.120.1	DC.DS01	DC.DS02
130	172.16.130.0/24	2	130	172.16.130.1	DC.DS02	DC.DS01
190	172.16.190.0/24	2	190	172.16.190.1	DC.DS02	DC.DS01

- Preempt function should be enabled on routers.

3.5. DHCP

VLAN ID	Device as DHCP Server	DHCP Configuration				
		Network	Default-router	DNS	Start IP	Number of IPs
210	OFF.DHCP01	172.17.210.0/24	172.17.210.1	118.69.79.79	172.17.210.10	200
220	OFF.DHCP01	172.17.220.0/24	172.17.220.1	118.69.79.79	172.17.220.10	200
230	OFF.DS01	172.17.230.0/24	172.17.230.1	118.69.79.79	172.17.230.10	200
810	BR.GW01	172.28.81.0/24	172.28.81.1	118.69.79.79	172.28.81.10	200
820	BR.GW01	172.28.82.0/24	172.28.82.1	118.69.79.79	172.28.82.10	200
830	BR.GW01	172.28.83.0/24	172.28.83.1	118.69.79.79	172.28.83.10	200

3.6. IP Address assignment

Host name	Interface	IP Address	Subnet mask	Default Gateway	DNS Server
HQ.GW01	S0/0/0	118.69.255.2	255.255.255.252	-	-
	S0/0/1	118.69.255.9	255.255.255.252	-	-
	Gi0/0	172.23.255.9	255.255.255.252	-	-
	Gi0/1	172.23.255.1	255.255.255.252	-	-
	Gi0/2	172.23.255.5	255.255.255.252	-	-
DC.DS01	Gi0/1	172.23.255.2	255.255.255.252	-	-
	VLAN 110	172.16.110.2	255.255.255.0	-	-
	VLAN 120	172.16.120.2	255.255.255.0	-	-
	VLAN 130	172.16.130.2	255.255.255.0	-	-
	VLAN 190	172.16.190.2	255.255.255.0	-	-
DC.DS02	Gi0/1	172.23.255.6	255.255.255.252	-	-
	VLAN 110	172.16.110.3	255.255.255.0	-	-
	VLAN 120	172.16.120.3	255.255.255.0	-	-

Host name	Interface	IP Address	Subnet mask	Default Gateway	DNS Server
	VLAN 130	172.16.130.3	255.255.255.0	-	-
	VLAN 190	172.16.190.3	255.255.255.0	-	-
OFF.DS01	Gi0/1	172.23.255.10	255.255.255.252	-	-
	Fa0/19	172.17.10.1	255.255.255.0	-	-
	VLAN 210	172.17.210.1	255.255.255.0	-	-
	VLAN 220	172.17.220.1	255.255.255.0	-	-
	VLAN 230	172.17.230.1	255.255.255.0	-	-
BR.GW01	S0/0/0	118.69.255.6	255.255.255.252	-	-
	S0/0/1	118.69.255.10	255.255.255.252	-	-
	Gi0/0.810	172.28.81.1	255.255.255.0	-	-
	Gi0/0.820	172.28.82.1	255.255.255.0	-	-
	Gi0/0.830	172.28.83.1	255.255.255.0	-	-
DC.ACC01	VLAN 190	172.16.190.11	255.255.255.0	172.16.190.1	-
DC.ACC02	VLAN 190	172.16.190.12	255.255.255.0	172.16.190.1	-
OFF.WLC01	Management	172.17.230.254	255.255.255.0	172.17.230.1	-
OFF.LAP-F1	Management	Dynamic assignment			
OFF.LAP-F2	Management	Dynamic assignment			
mb.com.vn	NIC0	172.16.110.4	255.255.255.0	172.16.110.1	118.69.79.79
APP1	NIC0	172.16.120.4	255.255.255.0	172.16.120.1	118.69.79.79
DB1	NIC0	172.16.130.4	255.255.255.0	172.16.130.1	118.69.79.79
DB2	NIC0	172.16.130.5	255.255.255.0	172.16.130.1	118.69.79.79
OFF.DHCP01	NIC0	172.17.10.200	255.255.255.0	172.17.10.1	118.69.79.79
Accountant1	NIC0	Dynamic assignment			
Counter1	NIC0	Dynamic assignment			
Guest1	WNIC0	Dynamic assignment			
Accountant2	NIC0	Dynamic assignment			
Counter2	NIC0	Dynamic assignment			
Guest2	WNIC0	Dynamic assignment			
Accountant3	NIC0	Dynamic assignment			
Counter3	NIC0	Dynamic assignment			
Guest3	WNIC0	Dynamic assignment			

3.7. Wireless LAN

Access Point	WLAN Network	WLAN Configuration			
		SSID	AP Group	Authentication	Passphrase
OFF.LAP-F1	172.17.230.0/24	HQ.Guest-F1	OFF.LAP-F1	WPA2-Personal	hq.mb@123
OFF.LAP-F2	172.17.230.0/24	HQ.Guest-F2	OFF.LAP-F2	WPA2-Personal	hq.mb@123
BR.AP01	172.28.83.0/24	BR.Guest	-	WPA2-Personal	br.mb@123

Notes: You are free to choose a PC to manage OFF.WLC01.

3.8. Credentials

Devices	Username	Pasword	Enable secret	Console password	VTY password
OFF.WLC01	admin	Cisco123	-	-	-
MB Bank's Routers/Switches	-	-	Cisco123	Cisco123	Cisco123

4. Routing and security requirements

4.1. Routing

- OSPF routing protocol is required to be enabled on MB Bank's routers and distributed L3 switches to propagate MB Bank's IP networks.
- Static default route is configured at HQ.GW01 router and this route is propagated to another routers/switches in MB Bank's network by OSPF.
- All MB Bank's LAN (*Head Quarter and Branch*) traffic should use primary link for Internet connection and use backup link only if the primary link is down.

4.2. Security requirements

- All routers and switches should be changed hostname according to Device configuration table.
- Routers and switches in MB Bank's Datacenter should be managed remotely through Telnet connections.
- All in-use access ports on access switches should be configured STP PortFast and STP BPDU Guard feature.
- All in-use access ports on access switches should be configured Port Security feature with following parameter:
 - o Maximum secured-MAC address: 3
 - o Violation mode: restrict.
 - o Dynamic secured-MAC address is sticked.
- All not in-use ports on all switches should be configured to be member of black hole VLAN and shut down.
- All trunking ports should be configured DTP non-negotiation.

5. For testing

- All end devices (PCs, laptops) can connect successfully to Internet (*fpt.vn at 118.69.79.78*) and servers in MB Bank's Datacenter with IPs or domain names.
- End devices can connect successfully each other.
- A failure of one of two MB Bank's DS switches in Datacenter doesn't impact network connectivity of servers.
- A failure of primary link doesn't impact Internet connectivity of MB Bank's users.